HILB^D OF A SMOOTH RELATIVE CURVE IS SMOOTH OF RELATIVE DIMENSION D

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 818-824 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 826-842, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_pic.json (pb=771, stmt 818-824, proof 826-842).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Author omits the verification of the relative dimension (fibre computation or dimension tracking) - named mechanical check; the induction and descent are in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 05B5: sources/stacks-pic/pinned-original-source.tex, source lines 842–842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B9G: sources/stacks-pic/pinned-original-source.tex, source lines 786–792. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VB: sources/stacks-pic/pinned-original-source.tex, source lines 842–842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VA: sources/stacks-pic/pinned-original-source.tex, source lines 842–842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $X \to S$ be a smooth morphism of schemes of relative dimension 1 such that the functors $\mathrm{Hilb}^d_{X/S}$ are representable. The schemes $\underline{\mathrm{Hilb}}^d_{X/S}$ are smooth over S of relative dimension d.*

Proof. We have $\underline{\mathrm{Hilb}}^0_{X/S} = S$ and $\underline{\mathrm{Hilb}}^1_{X/S} = X$ thus the result is true for $d = 0, 1$. Assuming the result for d, we see that $\underline{\mathrm{Hilb}}^d_{X/S} \times_S X$ is smooth over S (Morphisms, Lemma 01VB and 01VA). Since $\underline{\mathrm{Hilb}}^d_{X/S} \times_S X \to \underline{\mathrm{Hilb}}^{d+1}_{X/S}$ is finite locally free of degree $d + 1$ by Lemma 0B9G the result follows from Descent, Lemma 05B5. We omit the verification that the relative dimension is as claimed (you can do this by looking at fibres, or by keeping track of the dimensions in the argument above). □